Post Transformer Frontier AI Challenge

Post-Transformer Hackathon by Pathway | IIT Ropar | High Prep

Every transformer wakes up the same way. No memory of yesterday. No context from last week. Just weights frozen in time, processing each request as if it were the first conversation they've ever had, like Groundhog Day.

And the problems run deeper than forgetfulness:

- **Context that evaporates.** Ask a transformer to reason across a 100-page document, and watch it drown. The attention mechanism that makes transformers powerful becomes their ceiling. Memory that scales quadratically until it breaks.
- **Intelligence frozen in amber.** Your business evolves. Markets shift. Regulations change. But your model? Still thinking it's May 2025. Static training paradigms mean your AI can't learn from what happened yesterday, let alone adapt to what's happening now.
- **Memory that lives on sticky notes.** The industry's current "solution" to AI amnesia? Bolt on external databases. Vector stores. Retrieval pipelines. These aren't real memories. They're Post-it notes stuck to the outside of a brain that can't form its own. Your AI doesn't *internalize* knowledge. It doesn't build *contextualized* understanding. It just looks things up. There's a difference between remembering and searching.

These aren't engineering problems waiting for better optimization. They're fundamental architectural limitations. The transformer paradigm has a ceiling.

The Dragon Hatchling (BDH) breaks through it. In this High Prep contest, you’ll work with this state-of-the-art technology!

About Pathway and the Post-Transformer Era

Pathway builds the first post-transformer frontier model that solves AI's fundamental memory problem. While transformers wake up in the same state every time—like Groundhog Day—their architecture enables true continuous learning, infinite context reasoning, and real-time adaptation. They're not optimizing yesterday's technology; they're building what comes after transformers.

The company is led by co-founder and CEO **Zuzanna Stamirowska**, a complexity scientist who created a team of AI pioneers:

- **CTO Jan Chorowski** was among the first people to apply Attention to speech and worked with Nobel laureate Geoff Hinton at Google Brain.
- **CSO Adrian Kosowski** is a leading computer scientist and quantum physicist who obtained his PhD at age 20 and co-founded SPOJ, one of the earliest popular competitive programming platforms.

The company is backed by leading investors and advisors, including Lukasz Kaiser, co-author of the Transformer ("the T" in ChatGPT) and a key researcher behind OpenAI's reasoning models. Pathway is trusted by organizations such as NATO, La Poste, and Formula 1 racing teams. Pathway is headquartered in Palo Alto, California, with offices in Paris and Wrocław.

The Dragon Hatchling: A New Paradigm

The Dragon Hatchling thinks differently:

- **Memory that lives in the architecture.** Hebbian synapses that strengthen with experience, not external databases you query.
- **Context without ceilings.** O(T) linear attention that scales to infinite context without quadratic explosion.
- **Learning that never stops.** Architecture designed for adaptation, not frozen weights waiting for the next training run.
- **Reasoning you can see.** Sparse positive activations (roughly 5% of neurons) flow through interpretable graph structures.

Property	Transformer	BDH
Structure	Dense matrix layers	Scale-free graph of neurons
Activation	Nearly all neurons fire	Roughly 5% of neurons fire (sparse)
Memory	KV-cache (grows with context)	Hebbian synapses (constant size)
Attention	$O(T^2)$ complexity	O(T) linear complexity
Interpretability	Black box	Graph structure is directly visualizable

The paper was the #1 paper on Hugging Face the month it was published, #2 in 2025 despite being released in October, and trended for weeks. The research community has taken notice. Labs worldwide are exploring it.

But the paper is dense. The concepts might seem unfamiliar. Working with BDH requires understanding new primitives: scale-free graphs instead of dense matrices, Hebbian learning instead of static weights, and sparse activations instead of continuous signals.

This contest is your opportunity to work with a post-transformer breakthrough before the rest of the world catches up.

Your Mission – should you choose to accept it

This contest invites you to build tools that reveal how BDH works, showcase its unique capabilities, and demonstrate why it matters for the future of AI.

The best teams in this contest will be among the first to contribute to BDH's ecosystem. Your work could be featured by Pathway and shared with the global AI community. This isn't just an academic exercise—it's a chance to shape how the world understands post-transformer AI.

Three paths await you. Each offers a different way to contribute. Pick the one that matches your team's strengths.

Understanding BDH: Core Capabilities

Before building, understand what makes BDH fundamentally different. These aren't incremental improvements—they're architectural breakthroughs that transformers cannot replicate.

Architectural Strengths

1. **Constant-size memory regardless of context length**

   BDH maintains O(n×d) synaptic state no matter how long the sequence. While the transformer KV-cache grows linearly with every token (eventually exhausting GPU memory), BDH compresses history into a fixed-size Hebbian state. The architecture

guarantees this mathematically. Community experiments have demonstrated BDH running to 50k+ tokens with flat memory usage while transformers crash at ~12k on equivalent hardware.

2. **Native sparsity without tricks**

   Roughly 5% of neurons fire per token in BDH, compared to near-100% activation in transformers. This sparsity emerges naturally from the architecture—no L1 regularization, no pruning, no distillation required. Fewer active neurons means more efficient computation and, crucially, more interpretable behavior. When only 5% of neurons matter for a given input, you can actually trace what's happening.

3. **Monosemantic synapses by design**

   Individual synapses in BDH reliably strengthen for specific concepts. The paper demonstrates "currency synapses" that activate for currency names and "country synapses" for nations—consistent behavior across languages. This is built-in interpretability. Where transformer neurons are polysemantic (encoding multiple unrelated concepts), BDH synapses tend toward clean, single-concept encoding. You can point to a synapse and explain what it does.

4. **Inference-time learning via Hebbian updates**

   BDH doesn't just process tokens—it learns from them during inference. Synapses strengthen when neurons co-activate ("neurons that fire together wire together"). This means BDH can acquire new knowledge without backpropagation, without retraining, without fine-tuning. The model you deploy keeps getting smarter as it runs.

5. **Composable model merging**

   BDH's scale-free architecture enables concatenating independently trained models. Train a French translator and a Spanish translator separately, concatenate them, and get a model that handles both. This composability is architecturally native—not a post-hoc trick.

What This Means for Applications

These capabilities unlock use cases that transformers struggle with:

- **Healthcare:** Medical AI that accumulates patient context across visits without retraining or solves path-breaking reasoning problems that were previously not doable by AI.

Models that learn from new research papers at inference time. Interpretable reasoning chains with much better long-horizon reasoning capabilities.
- **Finance:** Fraud detection with continuous monitoring of money movement to uncover money laundering or financing of illicit activities.
- **Research:** AI assistants that build cumulative understanding of a research domain. Literature synthesis across thousands of papers. Interpretable hypothesis generation.

The question isn't whether BDH has these capabilities—the architecture guarantees them. The question is: **how does your project showcase them compellingly to the world?**

Showcase Opportunities for You

The following areas represent high-impact opportunities where compelling demonstrations would advance the field's understanding of post-transformer AI. These are directions where strong work can make a real contribution.

Visualization of Unique Properties

- No one has built the definitive BDH visualization. The Transformer Explainer from Georgia Tech became the standard way people understand transformers. BDH needs its equivalent—and the architecture is *more* visualizable, not less.
- **Why this matters:** BDH's sparse activations (5% vs 100%) create dramatic visual contrast. The scale-free graph topology is literally a graph you can render. Hebbian strengthening has temporal dynamics that transformers lack. A great visualization doesn't just explain BDH—it makes the advantages *visceral*.

Interpretability Demonstrations

- The paper shows qualitative examples of monosemantic synapses. A systematic demonstration—with clear methodology, compelling examples, and intuitive presentation—would be a significant contribution. The goal is not to "prove" monosemanticity (the paper already establishes this) but to showcase it in ways that resonate with researchers and practitioners.
- **Why this matters:** Interpretability is one of AI's biggest unsolved problems. BDH claims to solve it architecturally. Compelling demonstrations of this capability position BDH as the answer to a question the entire field is asking.

Long-Context and Continuous Learning Showcases

- BDH's constant memory enables reasoning at context lengths where transformers fail. Demonstrations showing BDH handling 50k, 100k, or longer contexts—especially on meaningful tasks—would be powerful evidence of the paradigm shift.
- Similarly, BDH's inference-time learning enables continuous adaptation. Demonstrations showing models that accumulate knowledge across sessions, learn from corrections, or adapt to distribution shift in real-time would showcase capabilities that transformers fundamentally lack.
- **Why this matters:** "Infinite context" and "continuous learning" are two of the most sought-after capabilities in AI. Concrete demonstrations matter more than theoretical claims.

Domain Applications

- BDH's unique properties map naturally to high-stakes domains where interpretability, long context, and continuous learning matter most.
- **Healthcare research:** Medical AI needs to reason across patient histories, integrate new research, and provide interpretable recommendations. BDH's architecture addresses all three.
- **Scientific discovery:** Research assistants who accumulate domain knowledge, synthesize across papers, and show their reasoning chains.
- **Enterprise AI:** Applications where auditability isn't optional—compliance, legal, and financial services.
- **Why this matters:** A working prototype in healthcare or scientific research demonstrates that post-transformer AI isn't theoretical—it's ready for the problems that matter most.

Three Paths to Choose From

Path A: Visualization and Inner Worlds

Make BDH's unique properties visible. Requires frontend skills AND understanding of BDH internals.

BDH is uniquely visualizable. Where transformers show opaque matrix multiplications, BDH shows sparse pulses of neural activity flowing through emergent graph structures.

Why BDH Visualization Is Different

What to visualize	Why it's possible in BDH	Impact
Sparse activations	Only ~5% of neurons fire. Dramatic contrast vs. dense networks.	Executives understand efficiency instantly. Researchers see which neurons matter.
Graph topology	Scale-free hub-and-spoke structure emerges spontaneously. Real graphs, not abstract matrices.	Shows how BDH organizes knowledge differently than transformers.
Hebbian learning	Synapses visibly strengthen when neurons co-activate. Temporal dynamics transformers lack.	Makes "neurons that fire together wire together" tangible.
Monosemantic synapses	Individual synapses respond consistently to specific concepts.	Interpretability you can point to and explain.

Project Directions

- **"Sparse Brain": Activation Density Comparator:** Side-by-side visualization showing BDH's ~5% activation vs. transformer's near-uniform activity. Same input, dramatically different neural behavior. Feed identical tokens to both architectures, render activation heatmaps, let users scrub through layers. The contrast should be visceral.
- **"Graph Brain": Emergent Topology Explorer:** Interactive force-directed graph of BDH's G_x = E @ D_x matrix. Show how random initialization evolves into brain-like hub structure during training. Click neurons to see what activates them. Filter by activation strength. Zoom into communities. Export subgraphs.
- **"Memory Formation": Hebbian Learning Animator:** Animate the σ matrix evolving as tokens are processed. Edge weights thicken when synapses strengthen. Time-lapse training to show which connections become permanent. "Neurons that fire together wire together" made visible across hours of training compressed into seconds.
- **"Pathfinder Live": Interactive Reasoning Demo:** Port the krychu/bdh pathfinding visualization to web. Users draw mazes, watch BDH reason through solutions layer by layer, see attention flow in real time. Compare to transformer solving the same maze. Show where BDH's approach differs.

Technical Requirements

Suggested team composition:

- At least one member in your Team with frontend expertise (JavaScript/TypeScript)
- Familiarity with D3.js, Svelte, or similar visualization libraries is strongly recommended
- For rapid prototyping: Streamlit/Gradio can produce working demos faster, but with less polish

Scope guidance:

Scope	Complexity	Notes
Static layer diagram	Low	Good starting point for any team
Sparsity comparison heatmap	Low-Medium	Achievable with basic D3/Plotly
Force-directed graph of topology	Medium	Benefits from D3.js experience
Animated Hebbian learning	Medium-High	Requires strong frontend skills
Full 3D walkthrough	High	Ambitious but achievable with Three.js

Judging Emphasis

Criterion	Weight
Visual clarity: Is it immediately understandable to non-experts?	30%
Technical correctness: Does it accurately represent BDH internals?	25%
Insight into architecture: Does it reveal something meaningful?	25%
Presentation quality: Demo polish, README, video	20%

Path B: Interpretability Showcases

Demonstrate BDH's interpretability advantages with rigor and clarity. For teams with ML research skills.

BDH claims built-in interpretability through monosemantic synapses and sparse activations. The paper establishes this conceptually. Your job: create compelling, systematic demonstrations that make these advantages tangible.

What Interpretability Means Here

Interpretability answers: "Why did BDH output X? Which synapses mattered? What concepts do they encode?"

This requires methodology. Clear experimental design. Reproducible results. Intuitive presentation. The goal is not to validate BDH's claims—those are established—but to showcase them in ways that advance understanding.

Project Directions

- **Monosemanticity Dashboard:** Systematically identify synapses that correlate with specific concepts. For each synapse, visualize activation patterns across input categories. Create an interactive explorer where users can query "what does synapse N encode?" and "which synapses activate for [concept]?" Make the abstract concrete.
- **Activation Atlas:** Map neuron firing patterns to input features. Cluster neurons by what activates them. Track how activation patterns shift during training. Build an explorer that makes BDH's internal organization legible.
- **Comparative Interpretability Study:** Apply the same interpretability methods to BDH and transformers. Concept probing on both. Activation analysis on both. Side-by-side comparison showing where BDH's native interpretability outperforms transformer black boxes.
- **Inference-Time Learning Demonstration:** Showcase BDH's ability to learn during inference. Design a clear protocol: teach the model new facts without backpropagation, demonstrate retention, show that transformers cannot do this. Make the "learning without training" capability undeniable.

Methodological Standards

Strong interpretability work requires:

- **Clear experimental design:** What are you measuring? How?
- **Reproducible methodology:** Others should be able to replicate your approach
- **Honest presentation:** Document limitations and future scope clearly
- **Intuitive visualization:** Numbers alone don't convince—show, don't just tell

Judging Emphasis

Criterion	Weight
Insight quality: Does this advance understanding of BDH's interpretability?	35%
Rigor: Is the methodology documented and reproducible?	30%
Communication: Clear to researchers and engineers?	25%
Community value: Would others build on this work?	10%

Path C: Open-Ended Frontier Exploration

Highest ambition, highest potential impact. For teams ready to push boundaries.

This path is deliberately open. Your goal: produce something that advances BDH's ecosystem in a direction no one has explored.

Possible Directions

Domain Applications

Build prototypes showing BDH in high-stakes domains:

- **Healthcare research:** Medical literature synthesis, patient history reasoning, interpretable clinical decision support

- **Scientific discovery:** Research assistants who accumulate domain knowledge and show reasoning chains
- **Legal/Compliance:** Document analysis at scale with auditable reasoning
- **Education:** Adaptive tutoring that learns student patterns without retraining

Pedagogical Contributions

- Build interactive explainers in the spirit of Transformer Explainer or CNN Explainer, but for BDH
- Create "build your own BDH from scratch" tutorials with runnable code
- Design a curriculum that takes someone from "what is attention?" to "how does Hebbian memory work?"

Ecosystem Infrastructure

- Better HuggingFace integration
- Evaluation harnesses for BDH-specific capabilities
- Benchmark suites for long-context and continuous learning
- Debugging and profiling tools

Novel Investigations

- Probe BDH behavior on tasks designed to showcase its strengths
- Explore architectural variations
- Investigate applications beyond language (time series, graphs, RL)

What Success Looks Like

A strong submission:

- Identifies a clear opportunity
- Executes with appropriate methodology
- Produces artifacts others can use or build on
- Documents limitations and future scope clearly

An exceptional submission:

- Creates something Pathway researchers would actually use
- Opens new application domains for BDH
- Produces work the community will reference

Judging Emphasis

Criterion	Weight
Novelty: Is this genuinely new territory?	30%
Rigor: Are claims supported by evidence, and are the engineering choices with the right hypotheses + research?	40%
Communication: Could others build on this?	30%

Resources

BDH Core Resources

Official Repository https://github.com/pathwaycom/bdh

The canonical BDH implementation. Clean PyTorch in bdh.py plus a training script based on Karpathy's nanoGPT scaffold. This is your starting point.

Key files:

- bdh.py: Core model architecture (RoPE, sparse activations, Hebbian state)
- train.py: Training loop for Tiny Shakespeare
- README.md: Setup instructions and architecture overview

Direct links:

- Model architecture: https://github.com/pathwaycom/bdh/blob/main/bdh.py
- Training script: https://github.com/pathwaycom/bdh/blob/main/train.py

Research Paper: https://arxiv.org/abs/2509.26507

"The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain"

HTML version (easier to navigate): https://arxiv.org/html/2509.26507v1

Official Company Page https://pathway.com

The main Pathway site features BDH prominently with the announcement "We're fundamentally changing the way models think."

Some Paper Sections

Paper Structure: https://arxiv.org/html/2509.26507v1. You can consider reading in this order.

Section	Title	Why Read It
Section 1	Introduction	Essential. Read the entire section deeply. Covers motivation (1.1), the key intuition combining modus ponens reasoning with Hebbian learning (1.2), contribution summary (1.3), and notation (1.4). This gives you the "why" behind everything.
Section 2	BDH: a language model architecture given by local distributed graph dynamics	The theoretical graph-based architecture. Section 2.2 defines "equations of reasoning" (local edge-reweighting). Section 2.5 explains the brain model interpretation with excitatory/inhibitory circuits.
Section 3	BDH-GPU: a tensor-friendly version of the BDH architecture	Critical for implementation. Shows how graph dynamics become GPU-efficient tensors. Section 3.2 defines BDH-GPU as a state-space system — Equation (8) is the core architecture. Section 3.4 shows equivalence between BDH and BDH-GPU.
Section 4	Implementation and scaling laws	Practical details. Section 4.1 covers implementation characteristics (layers, heads, sparse activation). Section 4.2 is the head-to-head GPT-2 comparison. Section 4.3 compares to other architectures (Linear Attention, Mamba, etc.).
Section 5	Analysis: emergence of modularity and scale-free structure	Important for visualization. Explains ReLU-lowrank blocks (5.2), signal propagation dynamics (5.3), and how modularity emerges (5.4). Section 5.5 has empirical findings on parameter distributions.

Section	Title	Why Read It
Section 6	Analysis: linear attention, sparse positive activation, and monosemanticity	The interpretability section. Section 6.3 covers monosemantic synapses with currency/country synapse examples. Section 6.4 covers the ~5% sparse neuron activations finding.
Section 7	Playing with the Hatchling	Experimental capabilities: model merging by concatenation (7.1), training without backpropagation through time (7.2).
Section 8	Conclusions	Takeaways for model engineering (8.1), implications for brain science (8.2), societal impact (8.3).
Appendix B	Further description of experiments	Details on translation task (B.1), scaling experiments (B.2), monosemantic synapse experiments (B.3), merging experiments (B.4).
Appendix E	BDH-GPU PyTorch code listing	Complete reference implementation. Cross-reference with bdh.py in the GitHub repo.

Reading path by goal:

- Conceptual understanding: Section 1 → Section 2 (especially 2.2, 2.5)
- Implementation: Section 3 (especially 3.2) → Section 4.1 → Appendix E
- Interpretability/Visualization: Section 5 → Section 6 (especially 6.3, 6.4)
- Experiments & Validation: Section 4.2 → Section 7 → Appendix B

For those familiar with current LLMs: Section 4.3 compares BDH to other architectures (Linear Attention variants, Mamba, etc.) — useful context if you know those models.

Video & Audio Deep Dives

Eye on AI Podcast — Post-Transformer Architectures & BDH
Explores the limitations of transformer-based models and introduces BDH as a post-transformer reasoning architecture, with discussion on memory, planning, and non-token reasoning.
YouTube: https://www.youtube.com/watch?v=E6WmXnEFDgc

Post-Transformer Era — Pathway Research Talk
Overview of the post-transformer paradigm, including reasoning systems beyond next-token prediction, BDH architecture intuition, and implications for real-world AI systems.
YouTube: https://www.youtube.com/watch?v=o9o7fU_ZSIE

SuperDataScience Podcast #929 — 72-minute technical deep dive
https://www.superdatascience.com/podcast/sds-929-dragon-hatchling-the-missing-link-between-transformers-and-the-brain-with-adrian-kosowski

Adrian Kosowski (CSO) with Jon Krohn. Start here for conceptual understanding before reading the paper. Covers intuition, motivation, sparse activations, and key architectural decisions.

The Neuron Podcast — "This AI Grows a Brain During Training" (48 min) Zuzanna Stamirowska (CEO) discusses BDH architecture, infinite context, live learning, interpretability, and real-world adopters (Formula 1, NATO, La Poste). Published January 7, 2026.

- YouTube: https://youtu.be/duw7RUif8hE
- Spotify: https://open.spotify.com/episode/1UtLR7jPyIzACw5oXp5SX4
- Apple Podcasts: https://podcasts.apple.com/us/podcast/this-ai-grows-a-brain-during-training-pathways-ai-w/id1742267001?i=1000744036292

EU-Startups Podcast Episode 146 — Zuzanna Stamirowska
https://www.eu-startups.com/2025/11/eu-startups-podcast-episode-146-zuzanna-stamirowska-co-founder-and-ceo-of-pathway/

The Futurists Podcast — ML in supply chain (45 min)
https://pathway.com/blog/futurist-podcast/

Written Reviews & Coverage

Major Publications

- The Wall Street Journal — "An AI Startup Looks Toward the Post-Transformer Era"
- The Wall Street Journal — "Tech That Will Change Your Life in 2026"
- Forbes — "Pathway Navigates Next Road For AI Foundational Models"
- Semafor — "New AI research claims to be getting closer to modeling human brain"

Technical Reviews & Analysis

- Paper review by Andrii Lukianenko — Detailed technical breakdown of architecture and brain model interpretation
- The Neuron Newsletter — Wait, Attention is NOT What You Need? — Deep analysis comparing BDH to DeepSeek's mHC and Grassmann
- Unite.AI Interview Series — Zuzanna Stamirowska
- Yahoo Finance — Pathway's Foundational LLM Collaboration with NVIDIA and AWS

Community Discussion

- **HuggingFace Papers** —
  - Community Q&A with authors https://huggingface.co/papers/2509.26507
  - #1 Paper of the Day and Month, #2 Paper of 2025
- **Alphaxiv** — Discussion and annotations https://alphaxiv.org/abs/2509.26507
- **EmergentMind** — Paper discussion https://www.emergentmind.com/papers/2509.26507

Community Showcases

These projects from recent hackathons demonstrate what's achievable. Study them to understand both the possibilities and where to push further.

1. **krychu/bdh** — Visualization Reference https://github.com/krychu/bdh

   Applies BDH to a pathfinding task and produces GIFs showing neuron dynamics, attention flow, and sparsity charts. Currently the best example of BDH visualization. Study how they instrumented the model, extracted activations, and rendered the

results. Key files: bdh.py (model), boardpath.py (training/inference/config), utils/ (visualization). Output examples: combined_board_neuron.gif, combined_attention_sparsity.gif.

2. **Pragadhishnitt** — Financial Reasoning + Interpretability Metrics https://github.com/Pragadhishnitt/bdh-pragadhishraajofficial

   NIT Trichy's team that applied BDH to S&P 500 earnings transcripts and sentiment analysis. Key contributions:

   - 14M parameter BDH outperforms 67M parameter DistilBERT on sentiment classification
   - Custom interpretability metrics: TMI (Topological Modularity), SPS (Synaptic Persistence), SEC (Sparsity-Entropy Correlation)
   - Two-stage protocol: Pre-training with regularization, then Hebbian inference

3. **Glass-brain** — Memory Scaling Demonstration https://github.com/sharmilcd/glass-brain

   This project by a CSE student at IIT Madras demonstrates O(1) memory scaling by running BDH to 50k+ tokens while the transformer crashes at ~12k on a T4 GPU. Includes:

   - Memory usage comparison graphs
   - Sparsity evolution heatmaps during training
   - Recurrent inference mode implementation

4. **Lig17/Track2** — Synaptic Scaffolding https://github.com/Lig17/Track2

   This theoretical contribution by final year undergrads implements inference-time learning via Hebbian synapses. Key concepts:

   - Fast weights (σ): Connections strengthen during inference
   - Metaplasticity (H): Frequently used synapses forget more slowly
   - Cross-session persistence: Synaptic state saved and reloaded

5. **takzen/vision-bdh** — Computer Vision Adaptation https://github.com/takzen/vision-bdh

   E-BDH configuration achieved 79.54% on CIFAR-10, outperforming ViT-Tiny (74%) with fewer parameters (3.2M vs 5.7M). ~35x faster training than initial implementation.

Alternative Implementations on GitHub

- **jploski/bdh-transformers**
  - HuggingFace compatible https://github.com/jploski/bdh-transformers
  - Use AutoModel, AutoConfig, and .generate() with BDH. Supports both recurrent and parallel attention modes.
- **severian42/BDH-MLX**
  - Apple Silicon optimized https://github.com/severian42/BDH-MLX
  - ~2x faster than PyTorch MPS on M-series Macs.
- **mosure/burn_dragon_hatchling**
  - Burn/Rust port https://github.com/mosure/burn_dragon_hatchling
- **adamskrodzki/bdh**
  - Dynamic vocabulary, stateful attention https://github.com/adamskrodzki/bdh

Visualization Inspiration

These visualizations set the standard for making neural architectures like Transformers accessible. The AI community at large needs equivalents for these for upcoming frontier work like BDH. Some examples:

Transformer Explainer (Georgia Tech)

- Demo: https://poloclub.github.io/transformer-explainer
- Code: https://github.com/poloclub/transformer-explainer
- Runs GPT-2 in browser using Svelte + D3.js + ONNX. Users type input, see attention patterns update in real time. This is the gold standard for architecture explainers.
- Possible Option For BDH (You can go beyond): Sparse activations and Hebbian updates offer more dramatic visual dynamics than attention matrices.

LLM Viz (Brendan Bycroft)

- Demo: https://bbycroft.net/llm
- Code: https://github.com/bbycroft/llm-viz

- 3D walkthrough of every operation in a transformer. Users navigate through the computation graph.
- Possible Option For BDH (You can go beyond)*:* Graph topology and sparse activations could make an even more compelling 3D visualization—fewer operations, more meaningful structure.

CNN Explainer (Georgia Tech)

- Demo: https://poloclub.github.io/cnn-explainer
- Code: https://github.com/poloclub/cnn-explainer
- Interactive convolutional network visualization. Click filters to see what they detect.
- Possible Option For BDH (You can go beyond)*:* The monosemantic synapse concept maps directly to "click to see what it encodes."

Colorful Vectors (HuggingFace Space)

- Space: https://huggingface.co/spaces/jphwang/colorful_vectors
- Embedding visualization through color mapping and interactive exploration.
- Possible Option For BDH (You can go beyond)*:* Neuron activation patterns could use similar dimensionality reduction approaches or interactive visualizations that enable easier understanding.

CraftGPT

- Video: https://www.youtube.com/watch?v=VaeI9YgE1o8
- Code: https://github.com/sammyuri/craftgpt
- Working LLM built in Minecraft redstone. Technical tour de force.
- Possible Option For BDH (You can go beyond)*:* BDH's graph structure might be easier to represent in unconventional mediums than transformer matrices.

Additional Learning Resources

- **nanoGPT** (BDH's implementation builds on this) https://github.com/karpathy/nanoGPT
- **Karpathy's "Let's build GPT" video** https://www.youtube.com/watch?v=kCc8FmEb1nY

Compute Resources

Free Tier Options:

Resource	GPU	VRAM	Limit	Best Use
Google Colab Free	T4	15GB	~12h session	Training small models, inference experiments
Kaggle Notebooks	T4/P100	16GB	30h/week	Longer training runs, batch experiments
HuggingFace Spaces	CPU or T4	Varies	Free tier	Hosting demos for submission
Lightning.ai	T4	15GB	22h/month	Overflow compute

Pay-as-you-go options:

- vast.ai — Marketplace for renting GPUs by the hour
- Lightning.ai — Cloud platform with GPU access
- Ola Krutrim Cloud — Indian cloud provider with GPU instances

Teams are encouraged to leverage additional resources available at their institutions.

Join the Community

Do upvote the paper: https://huggingface.co/papers/2509.26507 🙂

Star these repositories to stay updated:

- https://github.com/pathwaycom/pathway — The core Pathway engine
- https://github.com/pathwaycom/llm-app — LLM xPack and RAG/agent tooling
- https://github.com/pathwaycom/bdh — The Dragon Hatchling architecture

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Submission Requirements

A valid submission must include:

1. Hosted Demo (Optional)

Your solution must be accessible via web:

- HuggingFace Space: Free hosting, easy deployment
- Streamlit Cloud: Quick setup for Streamlit apps
- Gradio on HuggingFace: Good for interactive ML interfaces
- Custom website: If you have specific requirements

The demo must be functional at judging time. Ideally in the future too for the AI community to review it.

2. GitHub Repository

Note that you can develop your solution afterwards also, but create a github repository, in which you will make commits. Round 1 submission does not demand full implementation, but should have at least proper documentation of the proposed solution in README.

Public repository containing:

Code:

- Clean, documented implementation
- Requirements file (requirements.txt or pyproject.toml)
- Scripts to reproduce key results

README can include:

- What you built (one paragraph)
- What insight it reveals about BDH
- How to run locally
- How to access the hosted demo
- Team members and contributions
- Video demo and images
- **Limitations and future scope**

3. Demo Video

2-3 minute video showing:

- Your solution in action
- Key insights or findings
- Brief explanation of approach

Upload to YouTube (unlisted is fine) or include in the repository.

What Success Looks Like

A strong submission:

- Picks a clear, scoped problem
- Executes with appropriate methodology
- Produces a working demo and clean repository
- Documents limitations and future scope honestly
- Showcases something meaningful about BDH

A weak submission:

- Runs existing demos unchanged
- Builds visualizations disconnected from actual model behavior
- Surface-level engagement without genuine understanding
- Broken demo links or incomplete repository

An exceptional submission:

- Showcases a BDH capability in a way no one has before
- Creates artifacts Pathway researchers would reference
- Opens new application domains or research directions
- Sets a standard others will follow

Timeline

You have several weeks as this is a High Prep PS. Use them wisely.

Suggested Approach

Start with Foundation

- Read the paper (focus on Sections 2, 3, and 5)
- Watch the 72-minute video walkthrough between Adrian Kosowski and Jon Krohn
- Watch the podcast by Zuzanna Stamirowska on The Neuron
- Clone the repository, run the training script, understand the code
- Study the community showcases
- Pick your path and scope your project

Then Build →

- Implement core functionality
- Iterate based on what you learn
- Start documentation as you go

Finally Polish →

- Deploy to hosting platform
- Write comprehensive README (including limitations and future scope)
- Record demo video and create a presentation
- Final testing

BDH Frontier Challenge | Post-Transformer Hackathon by Pathway | IIT Ropar